

Febuton Film Coated Tablet 80mg (Febuxostat)

1. NAME AND STRENGTH OF ACTIVE INGREDIENT :

Each film coated tablet contains Febuxostat 80mg

2. LIST OF EXCIPIENTS:

Microcrystalline Cellulose, Lactose, Magnesium Stearate, Hydroxypropylmethyl Cellulose-L, Sodium Croscarmellose, Opadry

3. DESCRIPTION :

It occurs as an off-yellow to yellow, capsule shaped film-coated tablet, engraved with “980” on one surface and “S D” on the other side having a splitting line between S and D.

4. INDICATIONS :

Treatment of chronic hyperuricaemia in conditions where urate deposition has already occurred (including a history, or presence of, tophus and/or gouty arthritis).

Febuton is indicated in adults.

5. DOSAGE AND ADMINISTRATION :

The recommended oral dose of Febuton is 40 mg or 80 mg once daily without regard to food. The recommended starting dose of Febuton is 40 mg once daily. If serum uric acid is >6.0 mg/dL (357 µmol/L) after 2-4 weeks, Febuton 80 mg once daily may be considered. The 80 mg tablet can be divided into equal halves. In order to provide a 40 mg dose, the tablet should be split just before use. Prescribers should advise patients on how to break the tablets in half and to keep the other half for the next dose.

Febuton works sufficiently quickly to allow retesting of the serum uric acid after 2 weeks. The therapeutic target is to decrease and maintain serum uric acid below 6.0 mg/dL (357 µmol/L). Gout flare prophylaxis of at least 6 months is recommended (see Precautions).

Older People: No dose adjustment is required in the elderly (see Pharmacokinetics).

Renal Impairment: The efficacy and safety have not been fully evaluated in patients with severe renal impairment (creatinine clearance <30 mL/min, see Pharmacokinetics). No dose adjustment is necessary in patients with mild or moderate renal impairment.

Hepatic Impairment: The efficacy and safety of febuxostat has not been studied in patients with severe hepatic impairment (Child Pugh Class C).

No dose adjustment is necessary in patients with mild hepatic impairment. Limited information is available in patients with moderate hepatic impairment.

Paediatric Population: The safety and the efficacy of Febuton in children aged below the age of 18 years have not been established. No data are available.

Administration: Oral use.

Febuton should be taken by mouth and can be taken with or without food.

6. CONTRAINDICATIONS :

Hypersensitivity to the active substance or to any of the excipients listed (see List of excipients and Adverse Reactions).

7. PRECAUTIONS AND WARNINGS :

Cardiovascular Disorders: Treatment with febuxostat in patients with pre-existing major cardiovascular diseases (e.g. myocardial infarction, stroke or unstable angina) should be avoided, unless no other therapy options are appropriate. Caution should be exercised for exacerbation and/or onset of cardiovascular disease when administering this drug.

Medicinal Product Allergy/Hypersensitivity: Rare reports of serious allergic/hypersensitivity reactions, including life-threatening Stevens-Johnson Syndrome, Toxic epidermal necrolysis and acute anaphylactic reaction/shock, have been collected in the post-marketing experience. In most cases, these reactions occurred during the first month of therapy with febuxostat. Some, but not all of these patients reported renal impairment and/or previous hypersensitivity to allopurinol. Severe hypersensitivity reactions, including Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) were associated with fever, haematological, renal or hepatic involvement in some cases. Patients should be advised of the signs and symptoms and monitored closely for symptoms of allergic/hypersensitivity reactions (see Adverse Reactions). Febuxostat treatment should be immediately stopped if serious allergic/hypersensitivity reactions, including Stevens-Johnson Syndrome, occur since early withdrawal is associated with a better prognosis. If patient has developed allergic/hypersensitivity reactions including Stevens-Johnson Syndrome and acute anaphylactic reaction/shock, febuxostat must not be re-started in this patient at any time.

Acute Gouty Attacks (Gout Flare): Febuxostat treatment should not be started until an acute attack of gout has completely subsided. Gout flares may occur during initiation of treatment due to changing serum uric acid levels resulting in mobilization of urate from tissue deposits. At treatment initiation with febuxostat flare prophylaxis for at least 6 months with an NSAID or colchicine is recommended (see Dosage & Administration).

If a gout flare occurs during febuxostat treatment, it should not be discontinued. The gout flare should be managed concurrently as appropriate for the individual patient. Continuous treatment with febuxostat decreases frequency and intensity of gout flares.

Xanthine Deposition: In patients in whom the rate of urate formation is greatly increased (e.g.

malignant disease and its treatment, Lesch-Nyhan syndrome) the absolute concentration of xanthine in urine could, in rare cases, rise sufficiently to allow deposition in the urinary tract. As there has been no experience with febuxostat, its use in these populations is not recommended.

Mercaptopurine/Azathioprine: Febuxostat use is not recommended in patients concomitantly treated with mercaptopurine/azathioprine. Where the combination cannot be avoided patients should be closely monitored. A reduction of dosage of mercaptopurine or azathioprine is recommended in order to avoid possible haematological effects (see Interactions).

Organ Transplant Recipients: As there has been no experience in organ transplant recipients, the use of febuxostat in such patients is not recommended.

Theophylline: Co-administration of febuxostat 80 mg and theophylline 400mg single dose in healthy subjects showed absence of any pharmacokinetic interaction (see Interactions). Febuxostat 80 mg can be used in patients concomitantly treated with theophylline without risk of increasing theophylline plasma levels.

Liver Disorders: Mild liver function test abnormalities were observed in patients treated with febuxostat (5.0%). Liver function test is recommended prior to the initiation of therapy with febuxostat and periodically thereafter based on clinical judgment.

Thyroid Disorders: Increased TSH values (>5.5 $\mu\text{IU/mL}$) were observed in patients on long-term treatment with febuxostat (5.5%). Caution is required when febuxostat is used in patients with alteration of thyroid function.

Lactose: Febuxostat tablets contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Effects on Ability to Drive and Use Machines: Somnolence, dizziness, paraesthesia and blurred vision have been reported with the use of Febuxostat. Patients should exercise caution before driving, using machinery or participating in dangerous activities until they are reasonably certain that Febuton does not adversely affect performance.

8. DRUG INTERACTIONS:

Mercaptopurine/Azathioprine: On the basis of the mechanism of action of febuxostat on XO inhibition concomitant use is not recommended. Inhibition of XO by febuxostat may cause increased plasma concentrations of these drugs leading to toxicity (see Precautions). Drug interaction studies of febuxostat with drugs that are metabolized by XO have not been performed.

Drug interaction studies of febuxostat with cytotoxic chemotherapy have not been conducted. No data is available regarding the safety of febuxostat during cytotoxic therapy.

Rosiglitazone/CYP2C8 Substrates: Febuxostat was shown to be a weak inhibitor of CYP2C8 in vitro. Coadministration of 120 mg febuxostat QD with a single 4 mg oral dose of rosiglitazone had no effect on the pharmacokinetics of rosiglitazone and its metabolite N-desmethyl rosiglitazone, indicating that febuxostat is not a CYP2C8 enzyme inhibitor in vivo. Thus, co-administration of febuxostat with rosiglitazone or other CYP2C8 substrates is not expected to require any dose adjustment for those compounds.

Theophylline: The co-administration of febuxostat 80 mg QD with theophylline 400 mg single dose has no effect on the pharmacokinetics or safety of theophylline. Therefore no special caution is advised when febuxostat 80 mg and theophylline are given concomitantly.

Naproxen and Other Inhibitors of Glucuronidation: Febuxostat metabolism depends on Uridine Glucuronosyl Transferase (UGT) enzymes. Medicinal products that inhibit glucuronidation, such as NSAIDs and probenecid, could in theory affect the elimination of febuxostat. Concomitant use of febuxostat and naproxen 250 mg twice daily was associated with an increase in febuxostat exposure (C_{max} 28%, AUC 41% and $t_{1/2}$ 26%). The use of naproxen or other NSAIDs/Cox-2 inhibitors was not related to any clinically significant increase in adverse events.

Febuxostat can be co-administered with naproxen with no dose adjustment of febuxostat or naproxen being necessary.

Inducers of Glucuronidation: Potent inducers of UGT enzymes might possibly lead to increased metabolism and decreased efficacy of febuxostat. Monitoring of serum uric acid is therefore recommended 1-2 weeks after start of treatment with a potent inducer of glucuronidation. Conversely, cessation of treatment of an inducer might lead to increased plasma levels of febuxostat.

Colchicine/Indomethacin/Hydrochlorothiazide/Warfarin: Febuxostat can be co-administered with colchicine or indomethacin with no dose adjustment of febuxostat or the co-administered active substance being necessary.

No dose adjustment is necessary for febuxostat when administered with hydrochlorothiazide.

No dose adjustment is necessary for warfarin when administered with febuxostat.

Administration of febuxostat (80 mg or 120 mg once daily) with warfarin had no effect on the pharmacokinetics of warfarin in healthy subjects. INR and Factor VII activity were also not affected by the co-administration of febuxostat.

Desipramine/CYP2D6 Substrates: Febuxostat was shown to be a weak inhibitor of

CYP2D6 *in vitro*. 120 mg Febuton QD increase AUC of desipramine, a CYP2D6 substrate indicating a potential weak inhibitory effect of febuxostat on the CYP2D6 enzyme *in vivo*. Thus, co-administration of febuxostat with other CYP2D6 substrates is not expected to require any dose adjustment for those compounds.

Antacids: Concomitant ingestion of an antacid containing magnesium hydroxide and aluminium hydroxide has been shown to delay absorption of febuxostat (approximately 1 hour) and to cause a 32% decrease in C_{max}, but no significant change in AUC was observed. Therefore, febuxostat may be taken without regard to antacid use.

9. PREGNACY AND LACTATION

Pregnancy: Data on a very limited number of exposed pregnancies have not indicated any adverse effects of febuxostat on pregnancy or on the health of the foetus/new born child. The potential risk for human is unknown. Febuxostat should not be used during pregnancy.

Breastfeeding: It is unknown whether febuxostat is excreted in human breast milk. Febuxostat should not be used while breastfeeding.

10. ADVERSE REACTIONS :

Summary of the safety profile

The most commonly reported adverse reactions are gout flares, liver function abnormalities, diarrhoea, nausea, headache, rash and oedema. These adverse reactions were mostly mild or moderate in severity. Rare serious hypersensitivity reactions to febuxostat, some of which were associated to systemic symptoms, have occurred in the post-marketing experience.

Tabulated list of adverse reactions

Common, uncommon and rare adverse reactions occurring in patients treated with febuxostat are listed below.

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1: Adverse reactions

Blood and lymphatic system disorders	<u>Rare</u> Pancytopenia, thrombocytopenia
Immune system disorders	<u>Rare</u> Anaphylactic reaction, drug hypersensitivity
Endocrine disorders	<u>Uncommon</u> Blood thyroid stimulating hormone increased
Eye disorders	<u>Rare</u> Blurred vision
Metabolism and nutrition	<u>Common</u>

disorders	Gout flares <u>Uncommon</u> Diabetes mellitus, hyperlipidemia, decrease appetite, weight increase <u>Rare</u> Weight decrease, increase appetite, anorexia
Psychiatric disorders	<u>Uncommon</u> Libido decreased, insomnia <u>Rare</u> Nervousness
Nervous system disorders	<u>Common</u> Headache <u>Uncommon</u> Dizziness, paraesthesia, hemiparesis, somnolence, altered taste, hypoaesthesia, hyposmia
Ear and labyrinth disorders	<u>Rare</u> Tinnitus
Cardiac disorders	<u>Uncommon</u> Atrial fibrillation, palpitations, ECG abnormal
Vascular disorders	<u>Uncommon</u> Hypertension, flushing, hot flush
Respiratory system disorders	<u>Uncommon</u> Dyspnoea, bronchitis, upper respiratory tract infection, cough
Gastrointestinal disorders	<u>Common</u> Diarrhoea, nausea <u>Uncommon</u> Abdominal pain, abdominal distension, gastro-oesophageal reflux disease, vomiting, dry mouth, dyspepsia, constipation, frequent stools, flatulence, gastrointestinal discomfort <u>Rare</u> Pancreatitis, mouth ulceration
Hepato-biliary disorders	<u>Common</u> Liver function abnormalities <u>Uncommon</u> Cholelithiasis <u>Rare</u> Hepatitis, jaundice, liver injury
Skin and subcutaneous tissue disorders	<u>Common</u> Rash (including various types of rash reported with lower frequencies, see below)

	<u>Uncommon</u> Dermatitis, urticaria, pruritus, skin discolouration, skin lesion, petechiae, rash macular, rash maculopapular, rash papular <u>Rare</u> Toxic epidermal necrolysis, Stevens-Johnson Syndrome, angioedema, drug reaction with eosinophilia and systemic symptoms, generalized rash (serious), erythema, exfoliative rash, rash follicular, rash vesicular, rash pustular, rash pruritic, rash erythematous, rash morbilliform, alopecia, hyperhidrosis
Musculoskeletal and connective tissue disorders	<u>Uncommon</u> Arthralgia, arthritis, myalgia, musculoskeletal pain, muscle weakness, muscle spasm, muscle tightness, bursitis <u>Rare</u> Rhabdomyolysis, joint stiffness, musculoskeletal stiffness
Renal and urinary disorders	<u>Uncommon</u> Renal failure, nephrolithiasis, haematuria, pollakiuria, proteinuria <u>Rare</u> Tubulointerstitial nephritis, micturition urgency
Reproductive system and breast disorder	<u>Uncommon</u> Erectile dysfunction
General disorders and administration site conditions	<u>Common</u> Oedema <u>Uncommon</u> Fatigue, chest pain, chest discomfort <u>Rare</u> Thirst
Investigations	<u>Uncommon</u> Blood amylase increase, platelet count decrease, WBC decrease, lymphocyte count decrease, blood creatine increase, blood creatinine increase, haemoglobin decrease, blood urea increase, blood triglycerides increase, blood cholesterol increase, haematocrit decrease, blood lactate dehydrogenase increased, blood potassium increase <u>Rare</u> Blood glucose increase, activated partial thromboplastin time prolonged, red blood cell count decrease, blood alkaline phosphatase increase, blood creatine phosphokinase increase

Description of selected adverse reactions

Rare serious hypersensitivity reactions to febuxostat, including Stevens-Johnson Syndrome, Toxic epidermal necrolysis and anaphylactic reaction/shock, have occurred in the post-marketing experience. Stevens-Johnson Syndrome and Toxic epidermal necrolysis are

characterised by progressive skin rashes associated with blisters or mucosal lesions and eye irritation. Hypersensitivity reactions to febuxostat can be associated to the following symptoms: skin reactions characterised by infiltrated maculopapular eruption, generalised or exfoliative rashes, but also skin lesions, facial oedema, fever, haematologic abnormalities such as thrombocytopenia and eosinophilia, and single or multiple organ involvement (liver and kidney including tubulointerstitial nephritis) (see section warning and precaution).

Gout flares were commonly observed soon after the start of treatment and during the first months. Thereafter, the frequency of gout flare decreases in a time-dependent manner. Gout flare prophylaxis is recommended (see section dosage and administration and warning and precaution).

11. OVERDOSE AND TREATMENT :

Patients with an overdose should be managed by symptomatic and supportive care.

12. PHARMACOLOGY

Pharmacodynamics

Mechanism of Action:

Uric acid is the end product of purine metabolism in humans and is generated in the cascade of hypoxanthine → xanthine → uric acid. Both steps in the above transformations are catalyzed by xanthine oxidase (XO). Febuxostat is a 2-arylthiazole derivative that achieves its therapeutic effect of decreasing serum uric acid by selectively inhibiting XO. Febuxostat is a potent, non-purine selective inhibitor of XO (NP-SIXO) with an *in vitro* inhibition K_i value less than one nanomolar. Febuxostat has been shown to potently inhibit both the oxidized and reduced forms of XO. At therapeutic concentrations febuxostat does not inhibit other enzymes involved in purine or pyrimidine metabolism, namely, guanine deaminase, hypoxanthine guanine phosphoribosyltransferase, orotate phosphoribosyltransferase, orotidine monophosphate decarboxylase or purine nucleoside phosphorylase.

Pharmacokinetics

In healthy subjects, maximum plasma concentrations (C_{max}) and area under the plasma concentration time curve (AUC) of febuxostat increased in a dose proportional manner following single and multiple doses of 10 mg to 120 mg. For doses between 120 mg and 300 mg, a greater than dose proportional increase in AUC is observed for febuxostat. There is no appreciable accumulation when doses of 10 mg to 240 mg are administered every 24 hours. Febuxostat has an apparent mean terminal elimination half-life ($t_{1/2}$) of approximately 5 to 8 hours.

Absorption: Febuxostat is rapidly (t_{max} of 1.0-1.5 h) and well absorbed (at least 84%). After single or multiple oral 40 mg, 80 mg and 120 mg once daily doses, C_{max} is approximately 1.5-1.6 $\mu\text{g/mL}$, 2.8-3.2 $\mu\text{g/mL}$, and 5.0-5.3 $\mu\text{g/mL}$, respectively. Absolute bioavailability of

the febuxostat tablet formulation has not been studied.

Following multiple oral 80 mg once daily doses or a single 120 mg dose with a high fat meal, there was a 49% and 38% decrease in C_{max} and a 18% and 16% decrease in AUC, respectively. However, no clinically significant change in the percent decrease in serum uric acid concentration was observed where tested (80 mg multiple dose). Thus, Febuton may be taken without regard to food.

Distribution: The apparent steady state volume of distribution (V_{ss}/F) of febuxostat ranges from 29 to 75 L after oral doses of 10-300 mg. The plasma protein binding of febuxostat is approximately 99.2%, (primarily to albumin), and is constant over the concentration range achieved with 40 and 80 mg doses. Plasma protein binding of the active metabolites ranges from about 82% to 91%.

Biotransformation: Febuxostat is extensively metabolized by conjugation via uridine diphosphate glucuronosyltransferase (UDPGT) enzyme system and oxidation via the cytochrome P450 (CYP) system. Four pharmacologically active hydroxyl metabolites have been identified, of which three occur in plasma of humans. In vitro studies with human liver microsomes showed that those oxidative metabolites were formed primarily by CYP1A1, CYP1A2, CYP2C8 or CYP2C9 and febuxostat glucuronide was formed mainly by UGT 1A1, 1A8, and 1A9.

Elimination: Febuxostat is eliminated by both hepatic and renal pathways. Following an 80 mg oral dose of ¹⁴C-labeled febuxostat, approximately 49% of the dose was recovered in the urine as unchanged febuxostat (3%), the acyl glucuronide of the active substance (30%), its known oxidative metabolites and their conjugates (13%), and other unknown metabolites (3%). In addition to the urinary excretion, approximately 45% of the dose was recovered in the faeces as the unchanged febuxostat (12%), the acyl glucuronide of the active substance (1%), its known oxidative metabolites and their conjugates (25%), and other unknown metabolites (7%).

Renal Impairment: Following multiple doses of 80 mg of Febuton in patients with mild, moderate or severe renal impairment, the C_{max} of febuxostat did not change, relative to subjects with normal renal function. The mean total AUC of febuxostat increased by approximately 1.8-fold from 7.5 µg·h/mL in the normal renal function group to 13.2 µg·h/mL in the severe renal dysfunction group. The C_{max} and AUC of active metabolites increased up to 2- and 4-fold, respectively. However, no dose adjustment is necessary in patients with mild or moderate renal impairment.

Hepatic Impairment: Following multiple doses of 80 mg of Febuton in patients with mild (Child-Pugh Class A) or moderate (Child-Pugh Class B) hepatic impairment, the C_{max} and AUC of febuxostat and its metabolites did not change significantly compared to subjects with

normal hepatic function. No studies have been conducted in patients with severe hepatic impairment (Child-Pugh Class C).

Age: There were no significant changes observed in AUC of febuxostat or its metabolites following multiple oral doses of Febuton in elderly as compared to younger healthy subjects.

Gender: Following multiple oral doses of FEBURIC, the C_{max} and AUC were 24% and 12% higher in females than in males, respectively. However, weight-corrected C_{max} and AUC were similar between the genders. No dose adjustment is needed based on gender.

16. STORAGE CONDITIONS :

Store at or below 30°C.

17. DOSAGE FORMS AND PACKAGING AVAILABLE :

28 tablets /Box.

18. MANUFACTURED BY :

STANDARD CHEM & PHARM CO., LTD.

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